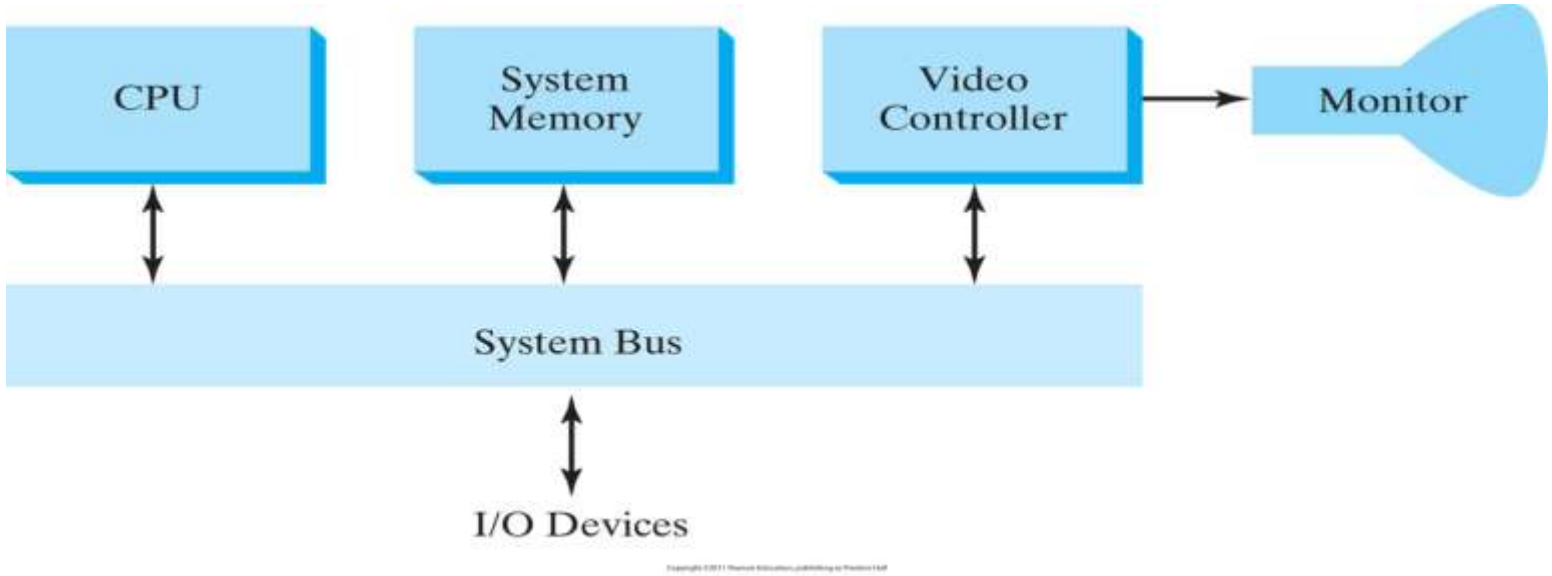
LESSON 3:
Graphics Display Technology

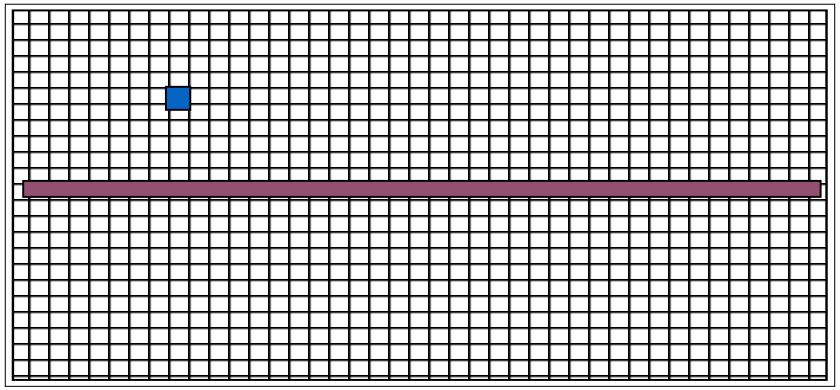
Basic Graphics Hardware

- Display devices
- Video controller
- Memory
- CPU
- System bus



Basic Definitions

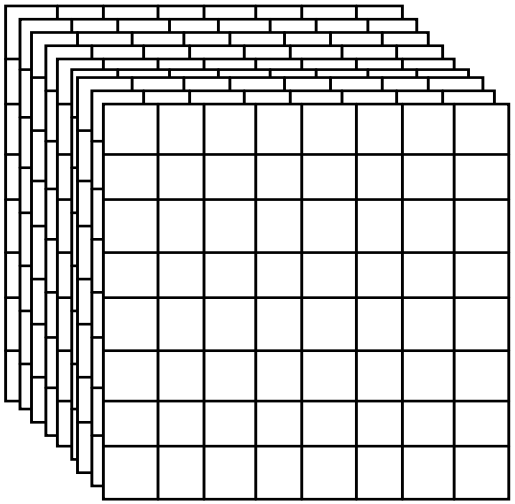
- Raster: A rectangular array of points or dots.
- Pixel (Pel): One dot or picture element of the raster
- Scan line: A row of pixels



Video raster devices display an image by sequentially drawing out the pixels of the scan lines that form the raster.

Frame Buffer

- A frame buffer is characterized by its size, x, y, and pixel depth.
- the **resolution** of a frame buffer is the number of pixels in the display. e.g. 1024x1024 pixels.
- Bit Planes or Bit Depth is the number of bits corresponding to each pixel. This determines the **color resolution** of the buffer.



Bilevel or **monochrome** displays have 1 bit/pixel (128Kbytes of RAM)
8bits/pixel -> 256 simultaneous colors
24bits/pixel -> 16 million simultaneous colors

Output Technology

- Calligraphic Displays
  - also called vector, stroke or line drawing graphics
  - lines drawn directly on phosphor
    - display processor directs electron beam according to list of lines defined in a "**display list**“
    - phosphors glow for only a few micro-seconds so lines must be redrawn or **refreshed** constantly
    - deflection speed limits number of lines that can be drawn without flicker.

Output Technology (2/3)

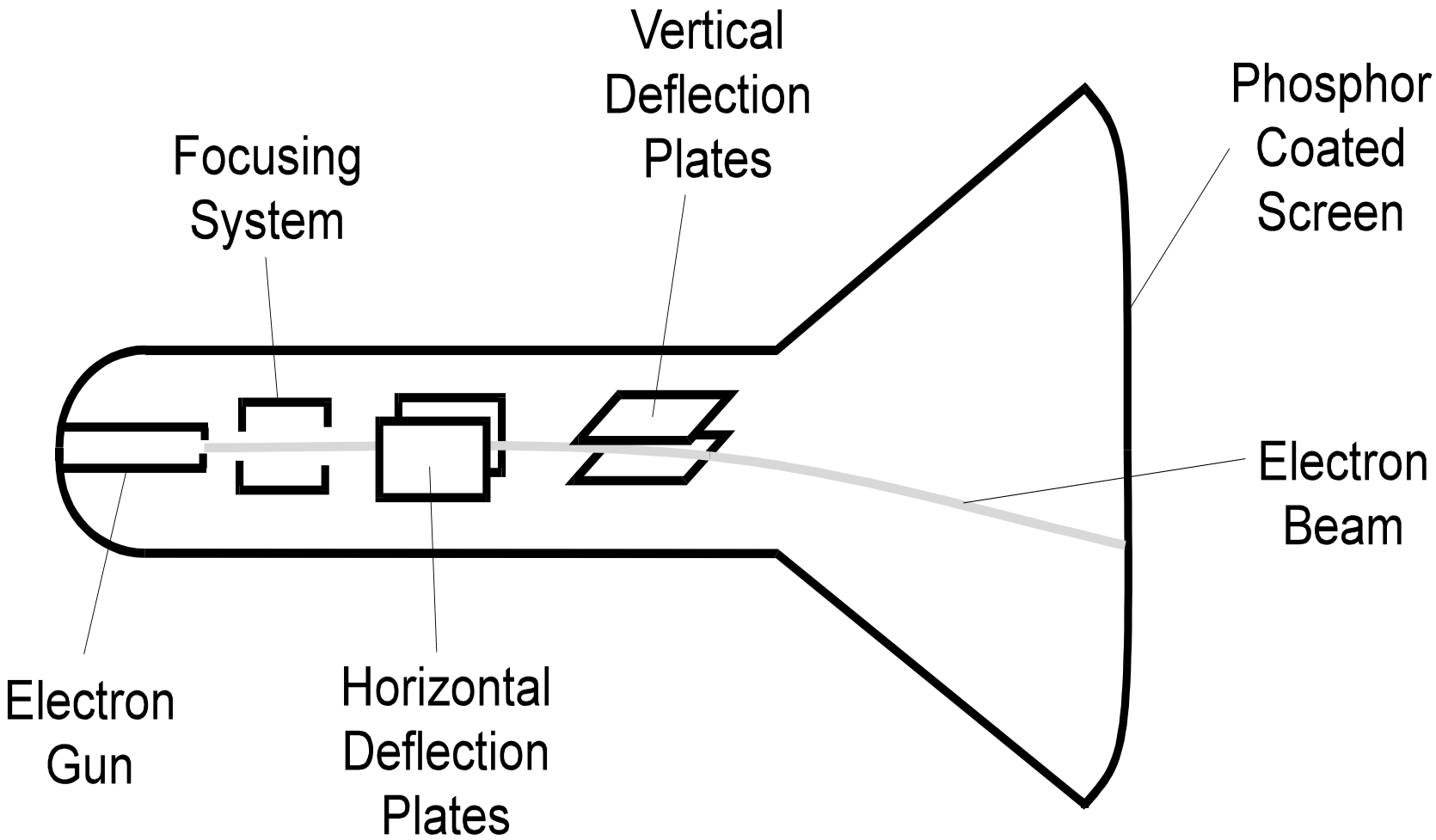
- Raster Display
  - Display **primitives** (lines, shaded regions, characters) stored as pixels in **refresh buffer** (or **frame buffer**)
  - Electron beam scans a regular pattern of horizontal raster lines connected by horizontal retraces and vertical retrace
  - Video controller coordinates the repeated scanning
  - Pixels are individual dots on a raster line

Display Technology

- 2D Displays
  - CRT
  - LCD (raster)
  - plasma screen (raster)
  - Light valves (raster)
  - Micromirror (raster)
  - Projected laser (vector)
  - Direct laser (vector)

- 3D Displays
  - Stereo presentation (raster/vector)
  - Vibrating mirror (vector)
  - Helical rotor (vector)
  - LED plate (raster)
  - Photoactive cube (raster)
  - Parabolic mirror (raster)

Display Technologies: Cathode Ray tube



Display Technologies

- Cathode Ray Tubes (CRTs)
  - Some of the common display device today
  - Evacuated glass bottle (last of the vacuum tubes)
  - Heating element (filament)
  - Electrons pulled towards anode focusing cylinder
  - Vertical and horizontal deflection plates
  - Beam strikes phosphor coating on front of tube

Display Technologies: CRTs

- Vector Displays
  - First computer displays: basically an oscilloscope
  - Control X,Y with vertical/horizontal plate voltage
  - Often used intensity as Z
  - *Disadvantage*
    - Display needs constant update to avoid fading

Display Technologies: CRTs

- Raster Displays
  - Black and white television: an oscilloscope with a fixed scan pattern: left to right, top to bottom
  - Paint entire screen 30 times/sec
    - Actually, TVs paint top-to-bottom 60 times/sec, alternating between even and odd *scanlines*
    - This is called *interlacing*.
  - To paint the screen, computer needs to synchronize with the scanning pattern of raster
    - Solution: special memory to buffer image with scan-out synchronous to the raster. We call this the *framebuffer*.

Comparing Raster and Vector

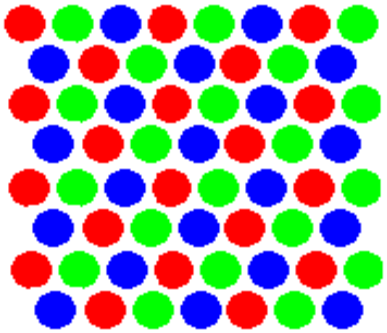
- Advantages of vector:
  - very fine detail of line drawings (sometimes curves), whereas raster suffers from jagged edge problem due to pixels.
  - geometry objects (lines) whereas raster only handles pixels
  - eg. 1000 line plot: vector dispaly computes 2000 endpoints
  - raster display computes all pixels on each line

Comparing Raster and Vector

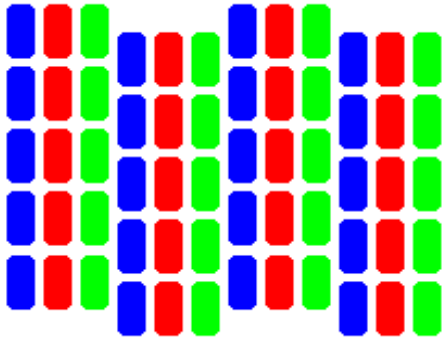
- Advantages of raster:
  - cheaper
  - colours, textures, realism
  - unlimited complexity of picture: whatever you put in refresh buffer, whereas vector complexity limited by refresh rate

Display Technology: Color CRTs

- Color CRTs are *much* more complicated
  - Requires manufacturing very precise geometry
  - Uses a pattern of color phosphors on the screen:



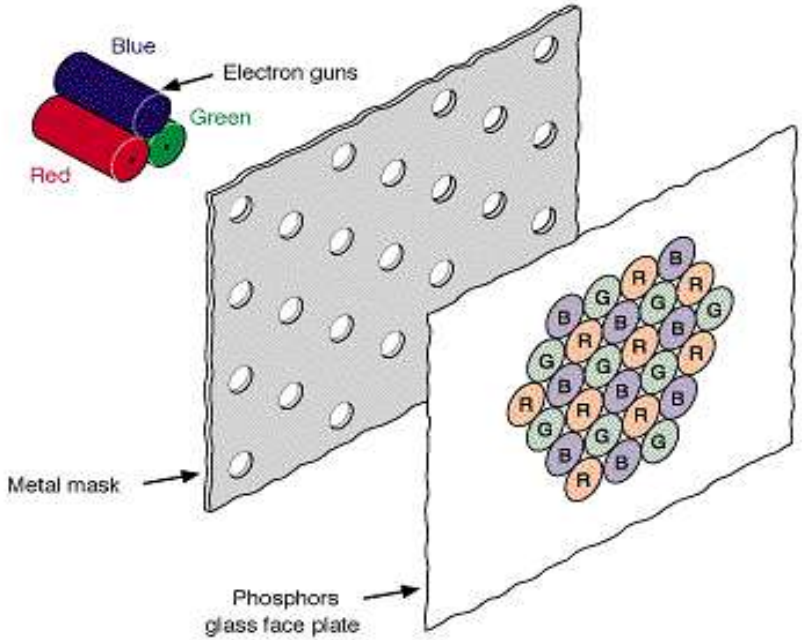
Delta electron gun arrangement

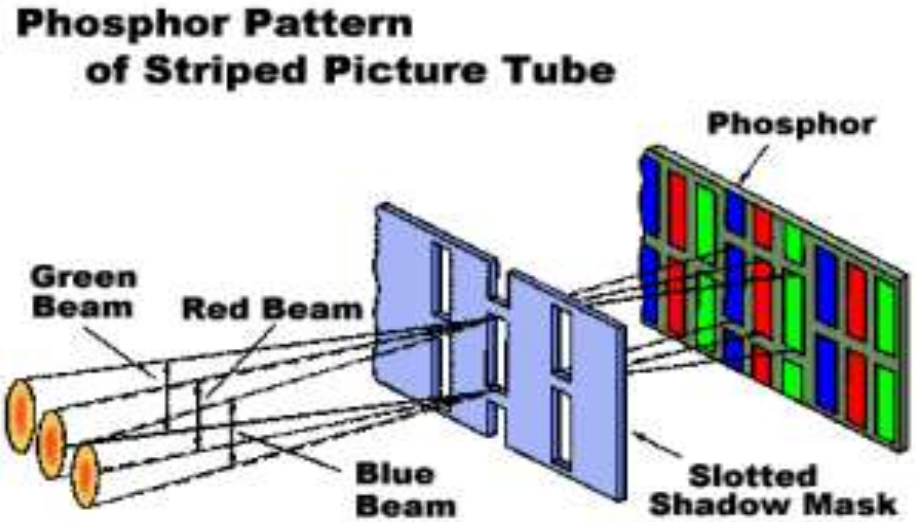


In-line electron gun arrangement

Display Technology: Color CRTs

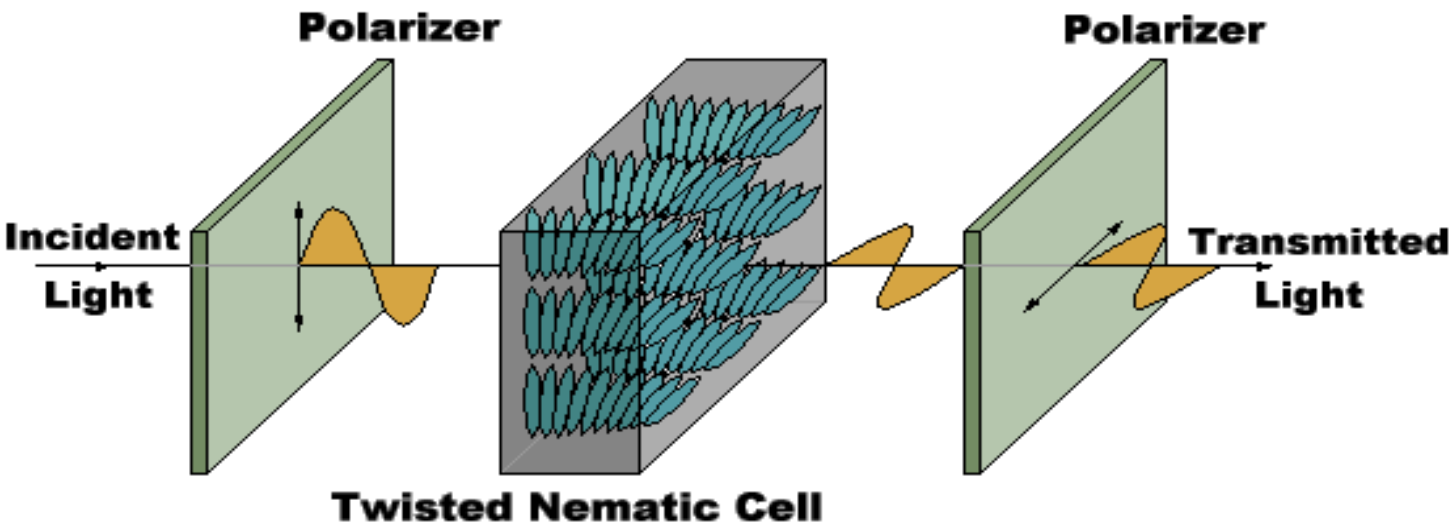
- Color CRTs have
  - Three electron guns
  - A metal *shadow mask* to differentiate the beams





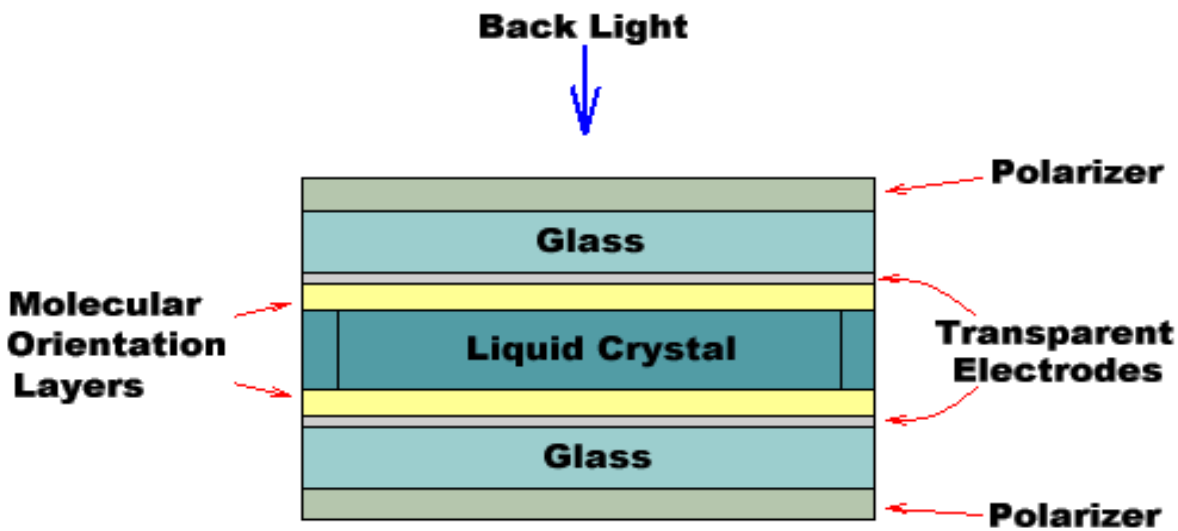
Display Technology: LCDs

- Liquid Crystal Displays (LCDs)
  - LCDs: organic molecules, naturally in crystalline state, that liquefy when excited by heat or E field
  - Crystalline state twists polarized light 90º.



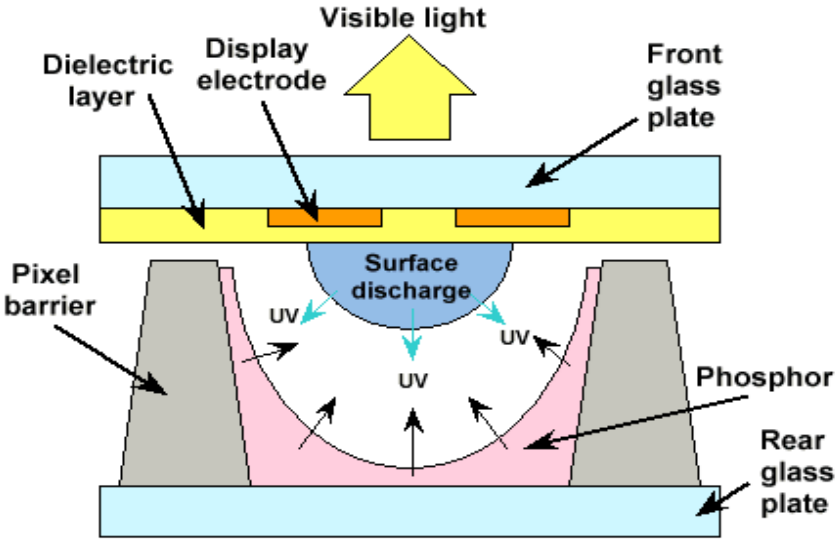
LCDs

- Transmissive & reflective LCDs:
  - LCDs act as light valves, not light emitters, and thus rely on an external light source.
  - Laptop screen: backlit, *transmissive display*
  - Palm Pilot/Game Boy: *reflective display*



Plasma

- Plasma display panels
  - Similar in principle to fluorescent light tubes
  - Small gas-filled capsules are excited by electric field, emits UV light
  - UV excites phosphor
  - Phosphor relaxes, emits some other color



Advantages of LCDs

- Physical Size
  - Compact and Lightweight
  - Space saving
  - Can be mounted on a wall or panel
- Display Size
  - Available at comparable in screen size as traditional CRT
  - LCD monitors are typically sized by their actual viewable diagonal measurement, but CRTs typically are not.
- Viewing
  - Cause less eyestrain
  - Does not flicker or glare

Advantages of LCDs

- Power Consumption and Radiation Emission
  - Doesn’t not emit Radiation
  - Not subject to Electromagnetic Interference
  - Consume less energy and more durable
    - A typical CRT losses approximately 50% of its brightness after 10,000 hours. An LCD bulb will maintain its brightness anywhere from 25,000 to 50,000 hours.
    - LCD consumes fewer watts than a CRT. LCD will use an average 30 watts compared to 120 watts for the CRT.
    - Can reduce electric bill by 40-85%.
    - Uses a combination of fluorescent-based backlight, color filters, transistors, and liquid crystal to create and illuminate images. It blocks light rather emit light

Advantages of CRT

- Color
  - Most are capable of displaying unlimited colors.
- Resolution
  - Multiple video Resolutions.
- Response Time
  - Faster response time. Critical to people who watch videos or play games on their PC's.
  - The fastest LCD's response time is about 25 milliseconds while a CRT's is about 13 milliseconds.
- Viewing Angle
  - Has a larger viewing angle

Disadvantages of LCDs

- Resolution
  - Displays Native Resolutions (Resolution that it displays best)
- Viewing Angle
  - Smaller, needed to be viewed more directly from the front.
  - From the side the images on an LCD screen can seem to disappear, or invert colors.
- Price
  - Upfront cost, it is more costly but will conserve energy in the long run.

Disadvantages of LCDs

- Installation
  - Need a plug interface to connect to the computer.
  - Some require a special digital plug-interface in order to work .
  - Problem: is that this plug is not available on most computers, so another video card or adapter must be purchased to plug these LCD monitors into the computer.
- Response Time
  - It is much slower. The delay can cause a ghosting effect on images it displays.

Disadvantages of CRTs

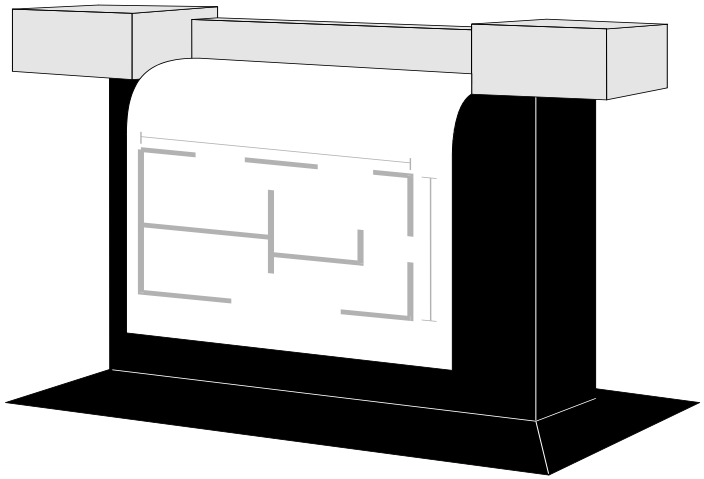
- Physical size
  - Takes more desktop space.
- Power Consumption and Radiation Emission
  - Consumes more energy
  - Emits harmful radiation. The metal shield behind the glass of your monitor protects your body from a flood of radiation.
  - It holds a dangerous electrical charge even after turned off and can last up to several years and be hazardous and even deadly to anyone who opens a CRT monitor casing.
  - Know to cause fires due to electrical malfunctions and excessive heat if the ventilation holes are accidentally covered up.

Input Devices

- Input devices
  - Keyboards, button boxes, dials
  - Mouse devices
  - Trackballs and space balls
  - Joysticks
  - Digitizers
  - Image scanners
  - Touch panels
  - Light pens
  - Voice systems
  - 3D Gloves

Hard Copy Devices

- Printers
  - Non-Impact printers --- Ink jet; laser;
  - Xerographic;
  - Electrostatic;
  - Dye sublimation.
- Plotters
  - Flatbed, Beltbed
  - Multiple pens available
  - Plotter `languages’
  - Built in character sets, line styles etc.



Pixels and Frames

- Images are created on television and computer displays by combining thousands of little dots (**pixels**) on the screen.
- Each set of pixels that fills the entire screen is called a **frame**.
- Many individual still frames are shown every second, so quickly that they blend together and we see continuous motion.
- The number of different frames that are shown on the screen every second relates to the **refresh rate**

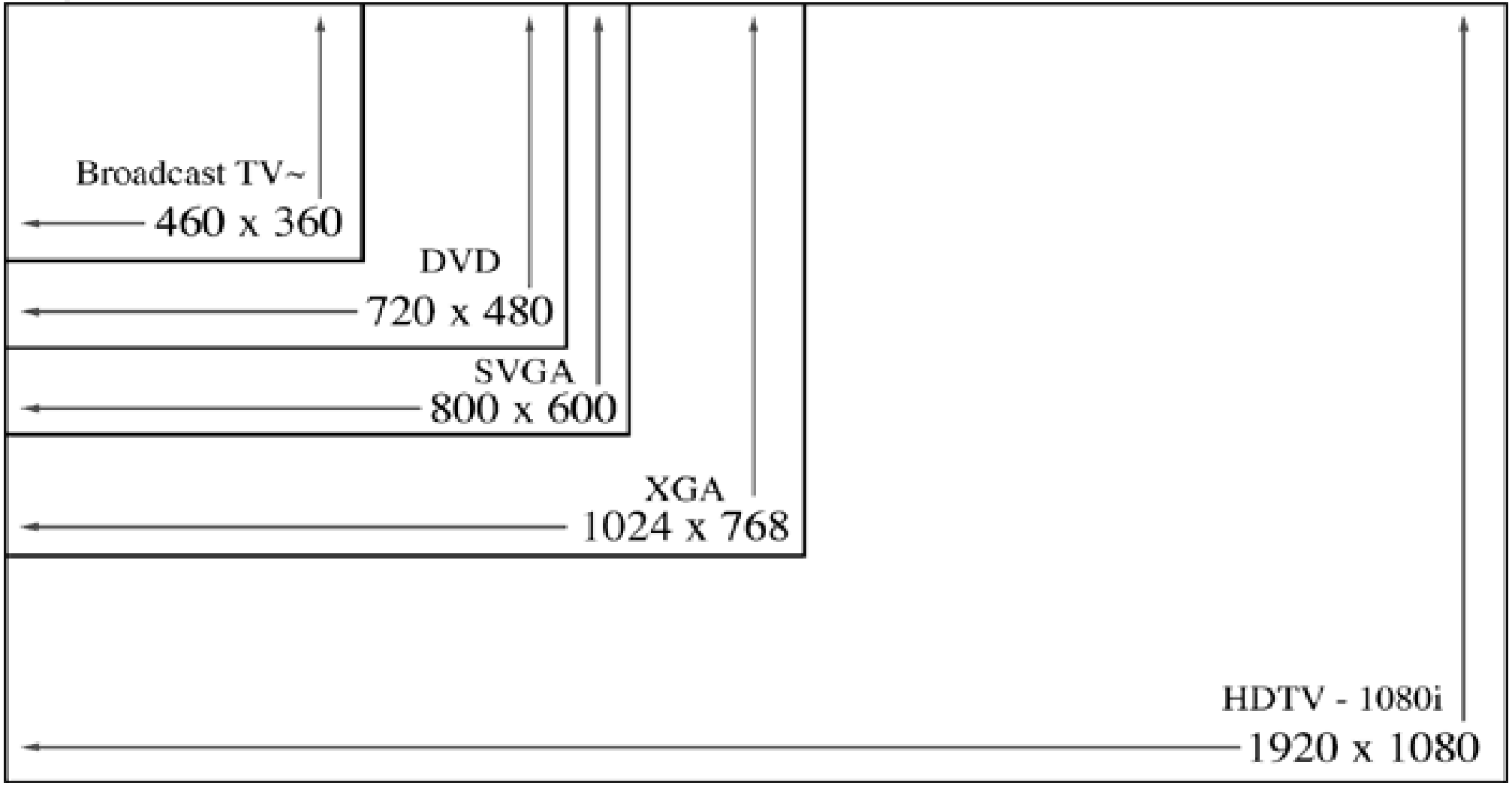
Pixels and Frames

- Each pixel in a color display is made up of three colors: red, green and blue.
- Red, green and blue light combines in varying intensities to allow each pixel to display all the colors of the rainbow and thousands or millions of shades in between.
- The number of shades of colors your computer can display relates to the **bit depth.**
- Each frame is drawn one pixel at a time, line by line, from left to right, top to bottom, just as you would type a paper

Resolution

- The number of pixels on a screen relates to display's **resolution**.
- The more pixels there are on a screen, the more detail can be seen in an image on the screen. **Higher resolution means smaller pixels**.
- If you have a **High-Definition Television**, or **HDTV**, then the display has a higher-resolution than a standard television.

Computer Display Resolution



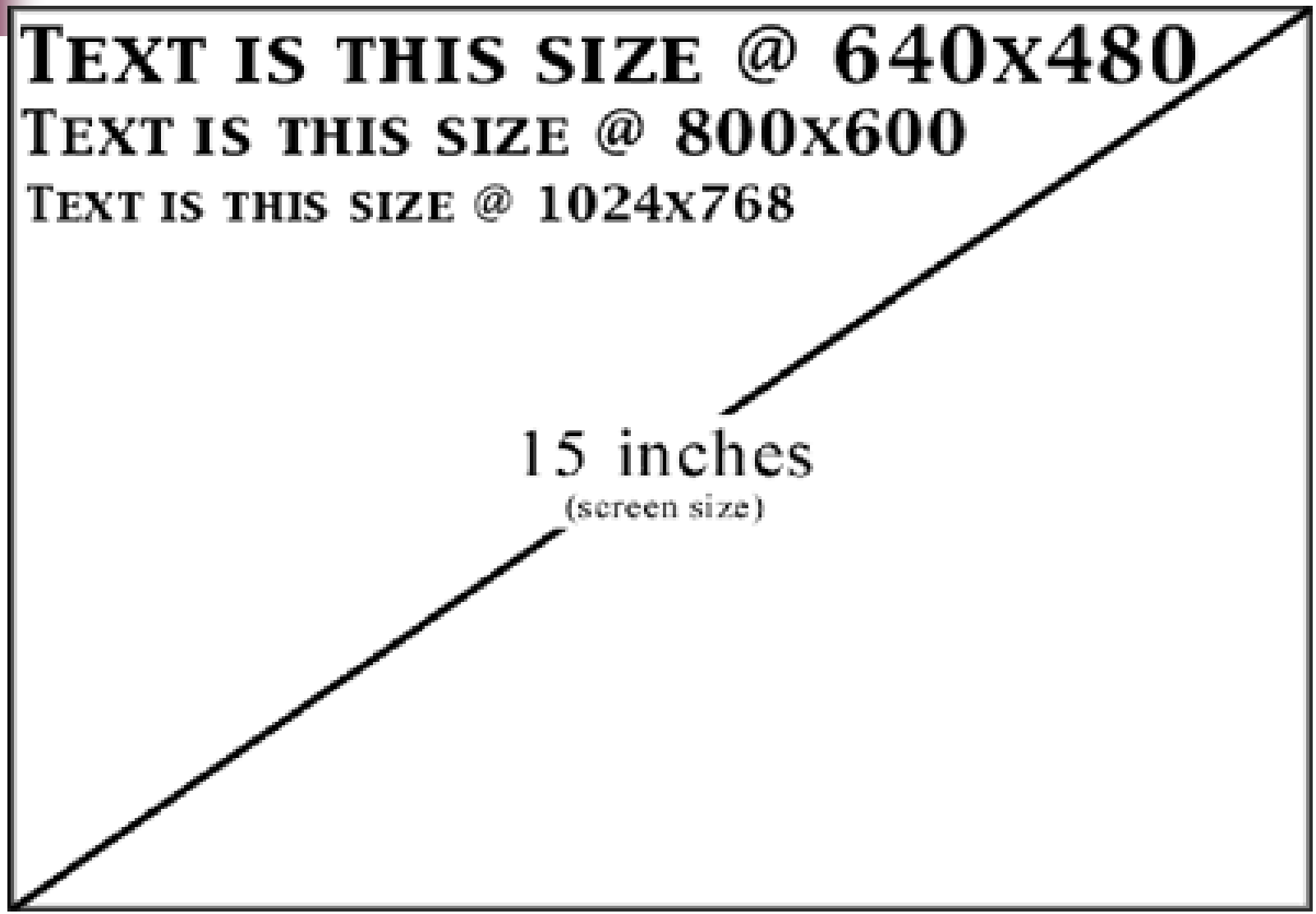
Computer Display Resolution vs. Pixel Size

- The three most common resolutions for computer displays are 640 x 480 (known as VGA), 800 x 600 (known as Super VGA) and 1024 x 768 (known as XGA).
- Most CRT displays can handle all of these resolutions, and many more.
- LCDs have a fixed number of pixels (usually at least 1024 x 768) and can use software to display resolutions below their fixed resolution with a process called **interpolation**, but cannot go above their fixed resolution.

Computer Display Resolution

- In higher resolution, the display must fit more pixels onto the screen , hence smaller pixels.
- An advantage of higher resolution is that you will be able to fit more toolbars onto your screen. The image below shows how a large font will shrink on your display when you switch to a higher resolution.
- Toolbars, menu bars and other objects on the screen will also shrink proportionally.

Computer Display Resolution



Refresh Rate and Frame Rate

- Pixels are drawn in rows across the screen from upper left to lower right and the entire display image, or frame, is updated numerous times per second.
- The term for "times per second" is **Hertz**, abbreviated **Hz**."
- The frequency at which the frames are displayed, or refreshed, is called the **Refresh Rate**.
- There are two ways to refresh the display: **interlaced** and **progressive scan**.

Refresh Rate and Frame Rate

- **Interlaced** means that every other line is drawn each time the display refreshes.
- For example lines 1, 3, 5, ... 457 and 459 are drawn in one frame, and lines 2, 4, 6, ... 458 and 460 are drawn in the next frame.
- The frames are drawn so quickly that they are blended together, or **interlaced**.

Refresh Rate and Frame Rate

- **Progressive scan** displays refresh every line of the frame each time the display refreshes.
- Computer monitors generally have variable refresh rates, with 60 Hz being the lowest.
- Displays can only refresh the frames up to a certain point, but our eyes do not notice any difference after about 85 Hz.
- A refresh rate of 85 Hz will give a good image without noticeable flicker to most peoples' eyes.

Refresh Rate and Frame Rate

- When we talk about movies or animation, we refer to the **frame rate**.
- The frame rate relates directly to the number of frames of the movie that are shown each second.
- Frame rates are given in units of fps (frames per second).
- For example, movies in the theater are shown at a frame rate of 24 fps.
- The refresh rate is always greater than or equal to the frame rate.

Bit Depth (Color)

- Bit depth relates to how many different colors each pixel can display.
- Computer monitors were originally monochrome - meaning that each pixel could only display two colors such as black and white.
- Monochrome monitors have now been replaced with RGB monitors.
- Each pixel on an RGB monitor contains three dots - one for each primary color.

Bit Depth (Color)

- RGB stands for the primary additive colors Red, Green and Blue, which combine to form every possible color.
- One bit color is what monochrome monitors use.
- A **bit** is a binary digit, which has two possible values - zero and one.
- A bit can represent colors, for example, black (0) and white (1).
- The more bits are used per pixel, the more colors it can display

Video Memory

- It used to be very simple to figure out how much video memory you needed.
- You just took the total number of pixels on the screen and multiplied that by the amount of memory that was used by each pixel.
- Nowadays, with 3-D graphics, texture rendering and other fancy stuff like that, computers use the video memory for a lot more than just displaying colored pixels.
- However, we can still use the formula for computers that do not have fancy video cards designed to handle 3-D graphics.

Video Memory

- Bits and Bytes represent a quantity of information, whether it be data **flowing** across a network or **stored** on a hard drive.
- There are eight bits in every byte.
- A bit is the tiniest unit of data.
- Bit can only have two values, zero or one.
- These values are easily represented in computers with electricity either off (0) or on (1).

How do bits and bytes relate to computer graphics?

- Computer graphics are nothing more than data, composed of little bits.
- Your computer needs enough storage in its video memory.
- **Q:** How do you calculate the amount of video memory a computer needs to display a given resolution at a given bit depth?
- **A:** The bit depth represents how many bits of memory each pixel uses.
- So, you multiply the number of pixels on your screen by the bit depth.
- To get the memory in bytes, just divide by eight (remember: 8 bits per byte).

Example 1

How much memory is required for an 800 by 600 display resolution with thousands of colors per pixel (16 bit)?

800 * 600 * (16 bits) / (8 bits / byte) = 960,000 bytes, or about 1 MB

You would need 1 MB of video memory to run your monitor at 800 by 600 with thousands of colors.

Example 2

Consider three different raster systems with resolutions of 640 x 480, 1280 x 1024, and 2560 x 2048.

i) What size is frame buffer (in bytes) for each of these systems to store 12 bits per pixel?

- Because eight bits constitute a byte, frame-buffer sizes of the systems are as follows:
- 640 x 480 x 12 bits / 8 = 450KB;
- 1280 x 1024 x 12 bits / 8 = 1920KB;
- 2560 x 2048 x 12 bits / 8 = 7680KB;

ii) How much storage (in bytes) is required for each system if 24 bits per pixel are to be stored?

- Similarly, each of the above results is just doubled for 24 (12×2) bits of storage per pixel.

Quiz

- Consider a raster system with the resolution of 1024 x 768 pixels and the color palette calls for 65,536 colors.
- What is the minimum amount of video RAM that the computer must have to support the above-mentioned resolution and number of colors?

Solution

- Recall that the color of each pixel on a display is represented with some number of bits. Hence, a display capable of showing up to 256 colors is using 8 bits per pixels (i.e., “8-bit color”).
- Notice, first that the color palette calls for 65,536 colors. This number is but 2^{16}, which implies that 16 bits are being used to represent the color of each pixel on the display.
- The display’s resolution is 1024 by 768 pixels, which implies that there is a total of 786,432 (1024 × 768) pixels on the display. Hence, the total number of bits required to display any of 65,536 colors on each of the screen’s 786,432 pixels is 12,582,912 (786,432 × 16).
- Dividing this value by 8 yields an answer of 1,572,864 bytes. Dividing that value by 1,024 yields an answer of 1,536 KB. Dividing that value by 1,024 yields an answer of 1.5 MB.

Example 3

- Consider two raster systems with the resolutions of 640 x 480 and 1280 x 1024.

i) How many pixels could be accessed per second in each of these systems by a display controller that refreshes the screen at a rate of 60 frames per second?

- Ans. Since 60 frames are refreshed per second and each frame consists of 640 x 480 pixels, the access rate of such a system is (640 x 480) * 60 = 1.8432×10^7 pixels/second.
- Likewise, for the 1280 x 1024 system, the access rate is (1280 x 1024) * 60 = 7.86432×10^7 pixels/second.

ii) What is the access time per pixel in each system?

- Ans. According to the definition of access rate, we know that the access time per pixel should be
- 1/(access rate). Therefore, the access time is around 54 nanoseconds/pixel for the 640 x 480 system, and the access time is around 12.7 nanoseconds/pixel for the 1280×1024 system.